

Module - IIUnits

Temperature - kelvin.

Pressure - Pa, (N/m^2 or kg/m^2) (Gauge or absolute).

Flow rate - kg/s .

Power - watt.

Electric power - kWh .

Lower heating value (LHV) - kJ/kg

Higher heating value (HHV) - kJ/kg

Standard cubic feet - scf (scuff)

scf $\Rightarrow$ cubic foot of gas at 60F & 14.73 psi.

Atmospheric pressure $\Rightarrow$ 760mm of Hg

$$1 \text{ atm} = 10^5 \text{ Pa}$$

1 $\text{kWh} \Rightarrow$ 1 kw power consumed in 1 hr.

HHV $\Rightarrow$ condensing & obtaining heat from water vapour which is a product of combustion process.

Petrol $\Rightarrow$ Mix of hydrocarbon, C_8H_{18}

Diesel Power Plant. (DPP)

- $\rightarrow$ DPP is a diesel engine coupled to a generator (2-50 MW).
- $\rightarrow$ Diesel or heavy fuel is used.

Application :

- Peak load plant
- mobile plant
- Standby units
- Emergency plant
- Starting stations
- : Central station...

- Peak load $\Rightarrow$ Max demand on power station $\Rightarrow$ low cost
- Base load $\Rightarrow$ Min demand. $\Rightarrow$ high η .
- Load curve $\Rightarrow$ Graphical representation b/w load in kW & time in hr. Shows variation of load on power station.

$$P = VI$$

when V is high, P is also high then, low I is needed.

Advantages

- Easy design & installation
- Available in diff capacity
- Response to load change is fast.
- Quick starting & stopping
- Less capital required.
- High η
- Installation near load centre.

Disadvantage

- High operating cost
- Pollution & noise
- High Increase in capacity is limited. (ltd. capacity).

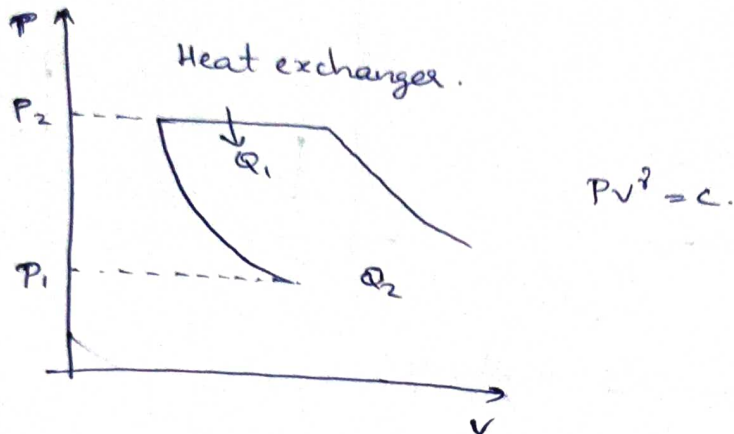
D.P.P. System

- Diesel engine
- Air intake system
- Exhaust system
- Fuel system
- Cooling system
- Lubrication system
- Starting system.

Gas Turbine Power plant.

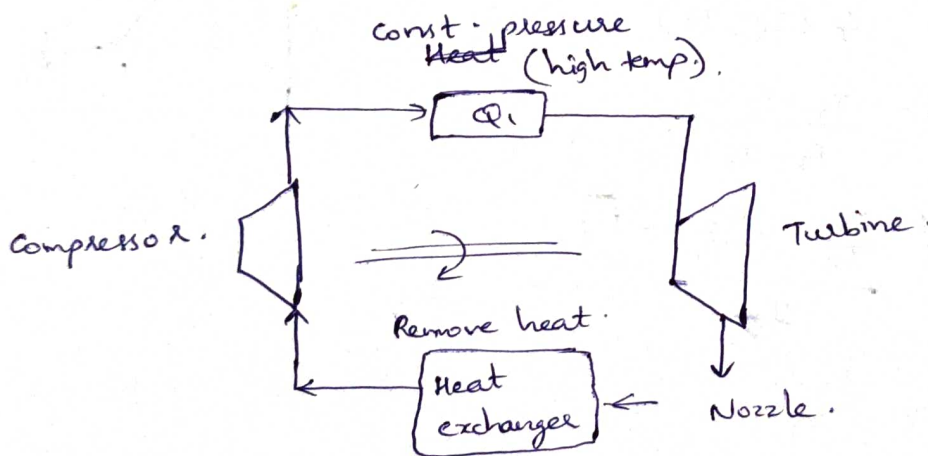
Gas turbine, Joule - Brayton cycle.

Brayton cycle (open cycle).

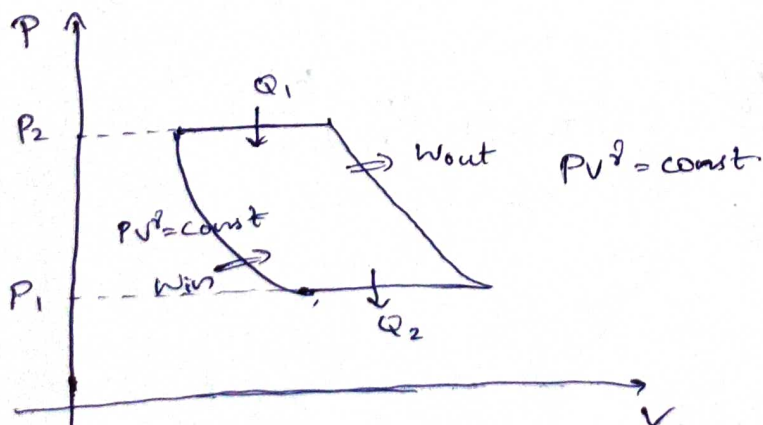


Heat exchanger $\Rightarrow$ device used to transfer heat b/w 2 fluids.

Gas turbine (closed)



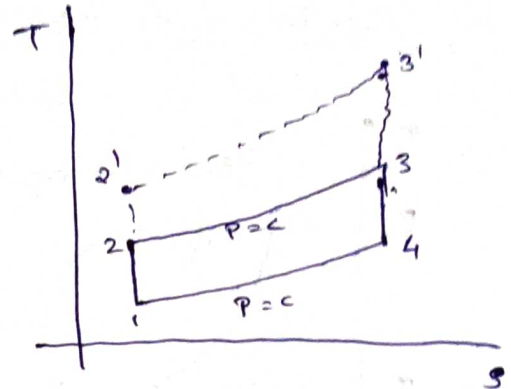
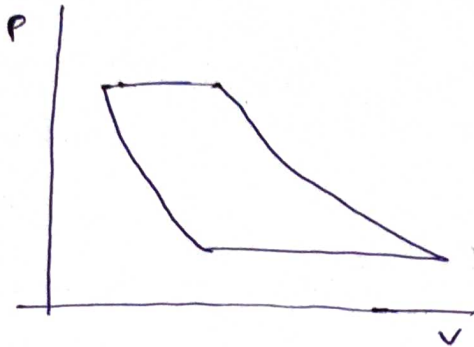
Joule - Brayton cycle.



Max temp = inlet temp to turbine.

It is limited $\because$ damage can occur.

$\downarrow$
depends on manufacturer.



Brayton Cycle analysis.

$$Q_{in} = c_p (T_3 - T_2)$$

$$Q_{out} = c_p (T_4 - T_1)$$

$$\eta = \frac{c_p (T_3 - T_2) - c_p (T_4 - T_1)}{c_p (T_3 - T_2)}$$

$$= 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

$$= 1 - \frac{T_1}{T_2} \left[\frac{T_4}{T_1} - 1 \right] \frac{1}{T_3}$$

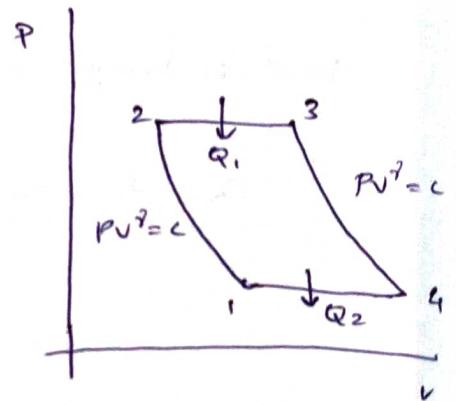
$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\therefore P_1 V_1^k = P_2 V_2^k \Rightarrow k = c_p / c_v$$

$$\frac{V_2}{V_1} = \left(\frac{P_1}{P_2} \right)^{1/k}$$

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{k-1}{k}}, \quad T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{k-1}{k}}$$

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{k-1}{k}}$$

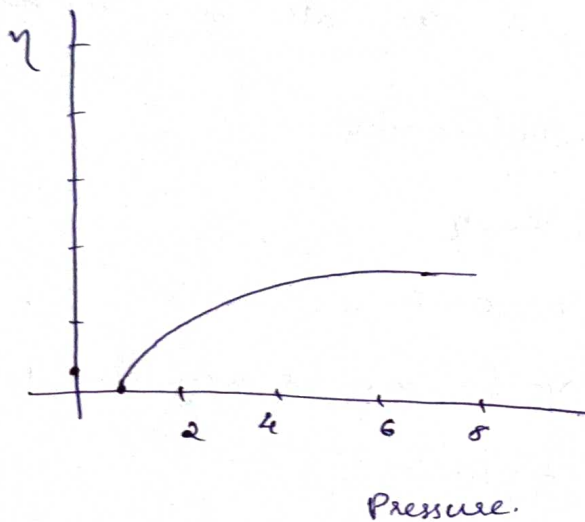


$$\frac{T_4}{T_3} = \left(\frac{P_4}{P_3} \right)^{\frac{k-1}{k}}$$

$$\frac{T_3}{T_4} = \frac{T_2}{T_1} \quad ; \quad \frac{T_3}{T_2} = \frac{T_4}{T_1}$$

$$\eta = \frac{1 - \frac{T_1}{T_2} \left[\frac{T_4}{T_1} - 1 \right]}{\frac{T_3}{T_2} - 1}$$

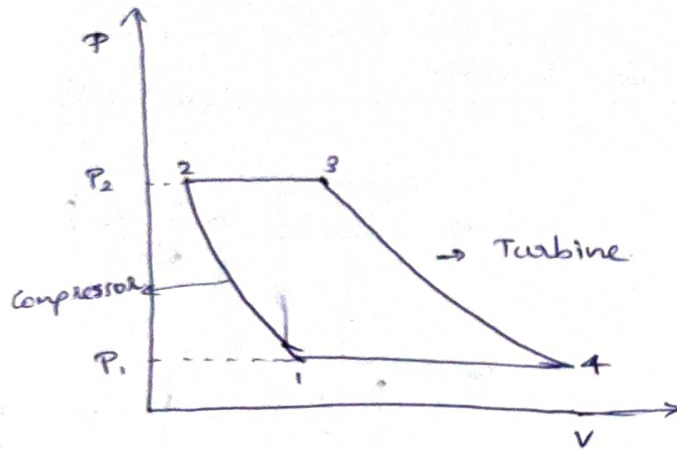
$$\eta = 1 - \frac{T_1}{T_2} = 1 - \frac{1}{\left(P_2/P_1 \right)^{\frac{k-1}{k}}}$$



→ 1 HP = 746 watt (most common conversion).

Power in watt.

Ideal Brayton Cycle.



Net work = Turbine work - Compressor work.

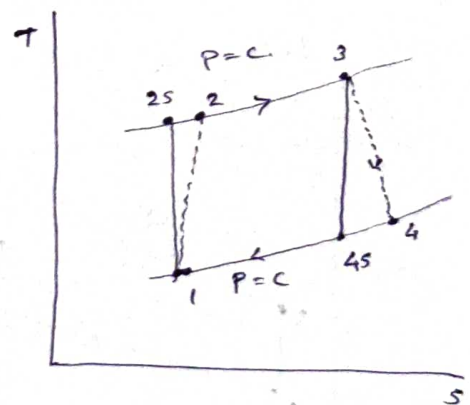
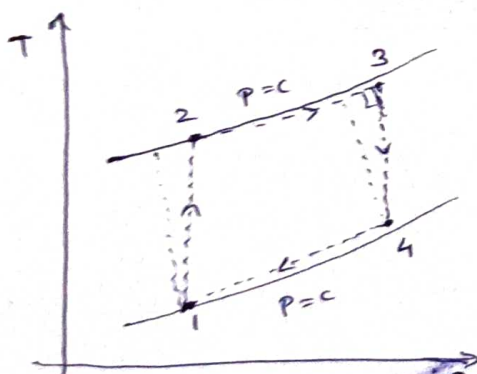
Optimum pressure ratio $\Rightarrow$

$$\frac{P_2}{P_1} = \left[\frac{T_3}{T_1} \right]^{\left[\frac{k}{2(k-1)} \right]}$$

where, $T_3 \Rightarrow$ max temp, $T_1 \Rightarrow$ min temp.

$$W_{net} = C_p (\sqrt{T_3} - \sqrt{T_1})^2 \Rightarrow \text{max work o/p.}$$

Gas turbine irreversibilities.



$$\eta_t = \frac{W_t / m}{(W_t / m)_s} = \frac{h_3 - h_4}{h_3 - h_{4s}} = \frac{T_3 - T_4}{T_3 - T_{4s}}$$

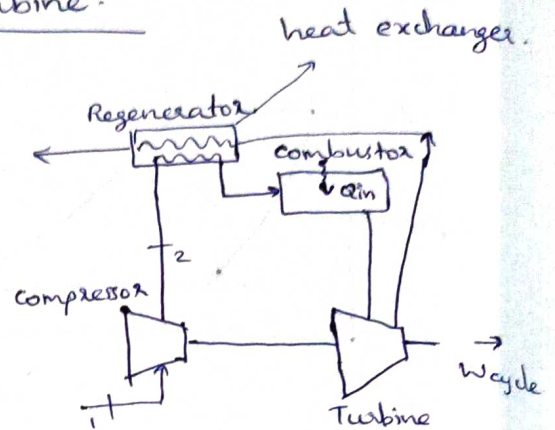
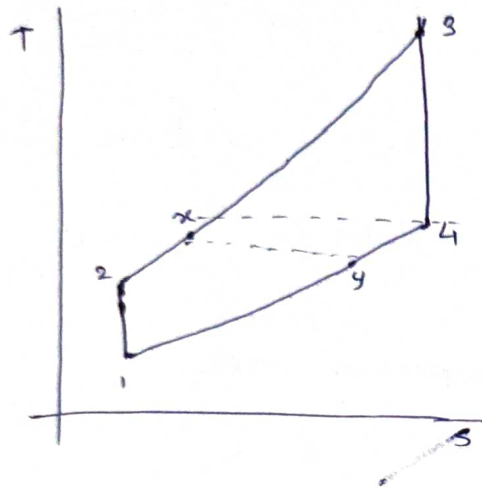
$$\eta_c = \frac{(W_c / m)_s}{W_c / m} = \frac{h_{2s} - h_1}{h_2 - h_1} = \frac{T_{2s} - T_1}{T_2 - T_1}$$

$\rightarrow$ Theoretical work o/p will be more than actual work o/p.

$$\eta_t = \frac{\text{act. w o/p}}{\text{Theoretical w o/p.}} \quad \text{for turbine} \rightarrow$$

In compressor, actual w.o/p will be high & theoretical w.o/p required will be less.

Regenerative Gas Turbine.

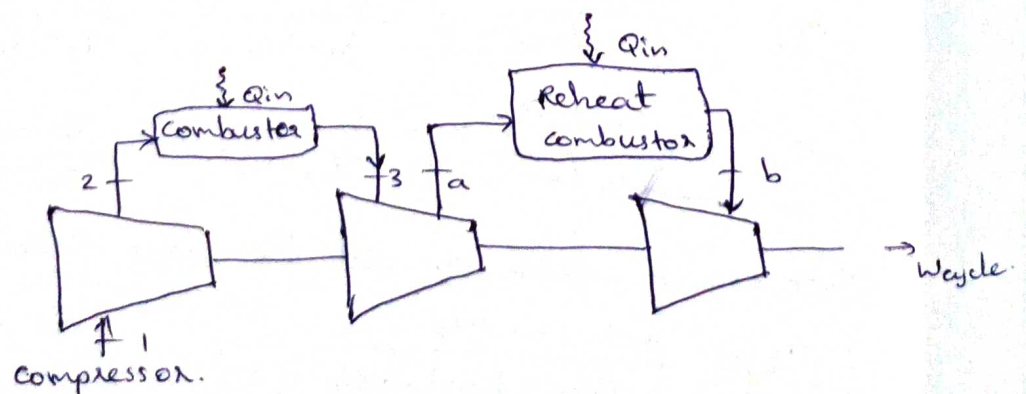
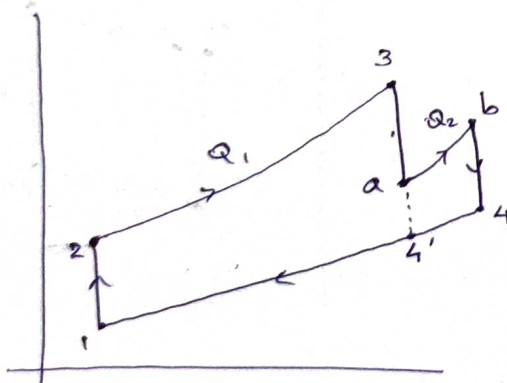


Usually heat addition takes place from 2-3 but $\therefore$ some heat is recovered, heat addition takes place from x-3.

Net work per unit mass $\uparrow$, $\eta_{th} \uparrow$.

$$\eta_{reg} = \frac{h_x - h_2}{h_4 - h_2}$$

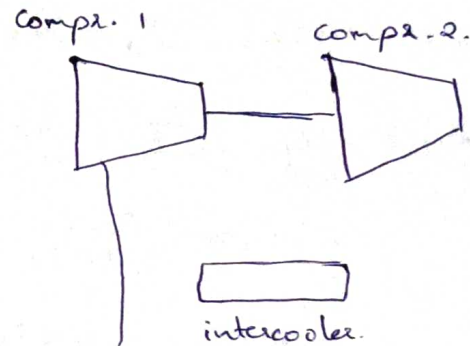
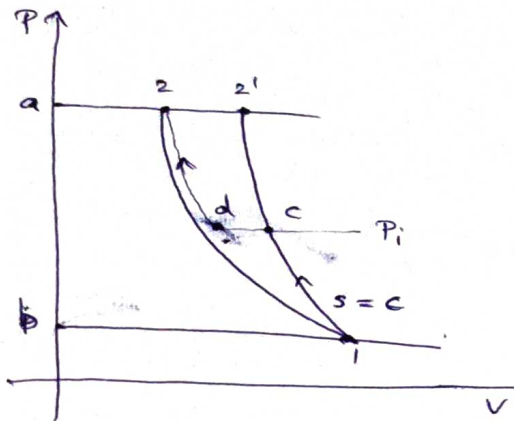
Reheat Gas Turbine



Efficiency doesn't change.

Compression with intercooling

Min temp $\Rightarrow$ min work.



During ~~with~~ increase compression, temp & pressure $\uparrow$. $\therefore$ more work is required. To reduce it we cool it (remove heat) $\therefore$ will compress again. So work i/p can be reduced.

$$\frac{W_c}{m} = (h_c - h_i) + (h_2 - h_d)$$

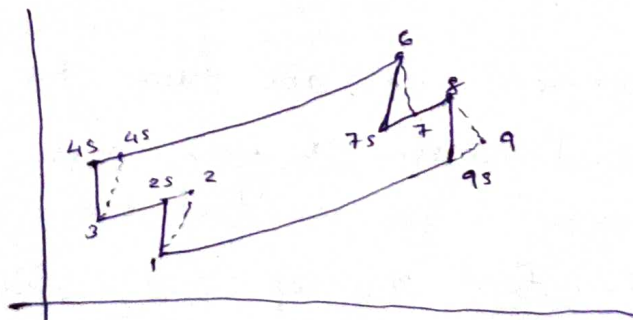
Assuming $T_i = T_d$ after intercooling;

$$\frac{W_c}{m} = c_p [(T_c - T_i) + (T_2 - T_i)]$$

Intercooler pressure for min compressor work,

$$P_i = \sqrt{P_1 P_2} \Rightarrow \text{intermediate } P \text{ for optimum work.}$$

$$\text{Optimum } P_i \Rightarrow \frac{\partial (W_c/m)}{\partial P_i} = \frac{c_p T_i}{P_i} \left(\frac{k-1}{k} \right) \left[\left(\frac{P_1}{P_i} \right)^{\frac{k-1}{k}} - \left(\frac{P_2}{P_i} \right)^{\frac{k-1}{k}} \right]$$



1. A gas turbine plant operates on the brayton cycle b/w min temp 300k & max temp 1073K. Find the max. w.d. per kg of air & corresponding cycle efficiency. How does this η compare with the carnot η . $c_p = 1.005 \text{ kJ/kgK}$

Ans. $T_3 = 1073\text{K}$, $T_1 = 300\text{K}$.

$$\begin{aligned} W_{\text{net}} &= c_p (\sqrt{T_3} - \sqrt{T_1})^2 \\ &= 1.005 (\sqrt{1073} - \sqrt{300})^2 \\ &= \underline{\underline{239.46 \text{ kJ/kg}}} \end{aligned}$$

$$\eta = 1 - \frac{1}{(P_2/P_1)^{\frac{k-1}{k}}} = 1 - \sqrt{\frac{T_1}{T_3}}$$

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}, \quad \frac{T_4}{T_3} = \left(P_1/P_2\right)^{k-1/k}$$

Assume, $k = 1.4$

$$\frac{1073}{300} = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}} = 3.57$$

$$\therefore \eta = 1 - \frac{1}{3.57} = \underline{\underline{72.04\%}}$$

$$\eta_{\text{br.}} = 1 - \sqrt{\frac{T_1}{T_3}} = \underline{\underline{47\%}}$$

$$\eta_{\text{car}} = \frac{T_1 - T_2}{T_1} = \underline{\underline{72.04\%}}$$

2. In an ideal brayton cycle, air from the atm at 1atm 300k is compressed 6atm & max cycle temp is limited to 1100K by using a large air fuel ratio. If the heat supplied is 100MW, find η_{th} of cycle, work ratio, power of

per kg of air. Assume, $c_p = 1.005 \text{ kJ/kg K}$, & ratio of specific heat 1.4.

Ans. $P_1 = 1 \text{ atm} = 10^5 \text{ Pa}$. $T_1 = 300 \text{ K}$.

$$P_2 = 6 \text{ atm} = 6 \times 10^5 \text{ Pa}. \quad T_3 = 1100 \text{ K}$$

$$Q_{in} = 100 \times 10^6 \text{ W}.$$

$$\eta = 1 - \frac{1}{\left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}} = 1 - \frac{1}{\left(\frac{6 \times 10^5}{1 \times 10^5}\right)^{0.28}} = \underline{\underline{39.4\%}}$$

$$\eta = \underline{\underline{40\%}}$$

$$\text{Pressure ratio, } r_p = \frac{P_2}{P_1}$$

$$\text{Specific heat ratio, } k = \frac{k-1}{k}$$

$$W_t = c_p (T_3 - T_4)$$

$$W_c = c_p (T_2 - T_1)$$

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}} \Rightarrow \frac{T_2}{T_1} = \left(\frac{6}{1}\right)^{\frac{0.4}{1.4}} = 1.66$$

$$\therefore T_2 = T_1 \times 1.66 = \underline{\underline{500.5 \text{ K}}}$$

$$\frac{T_4}{T_3} = \left(\frac{P_1}{P_2}\right)^{\frac{k-1}{k}} \Rightarrow T_4 = T_3 \left(\frac{P_1}{P_2}\right)^{\frac{k-1}{k}}$$

$$\therefore T_4 = 1100 \times \left(\frac{1}{6}\right)^{\frac{0.4}{1.4}} = \underline{\underline{659.27 \text{ K}}}$$

$$W_t = 1.005 \times (1100 - 659.27) = \underline{\underline{442.9 \text{ kJ/kg}}}$$

$$W_c = 1.005 \times (500.5 - 300) = \underline{\underline{201.5 \text{ kJ/kg}}}$$

$$\text{Work ratio} = \frac{W_t - W_c}{W_t} = \underline{\underline{0.54}}$$

$$\text{Power output} = \eta \times Q_{in} = 0.40 \times 100 = \underline{\underline{40 \text{ MW}}}$$

↓
(work o/p).

$$\frac{W}{Q_{in}} = \eta$$

3. The following refer to a stationary gas turbine, compressor inlet temp 311 K & compressor pressure ratio 8, compression chamber pressure drop 5% of inlet pressure, turbine inlet temp 1367 K. turbine exit & inlet compressor pressures are atmospheric. There exist a facility to take air from compressor exit for use in the cooling of the turbine. Find % of air that may be taken from the compressor for the purpose so that overall cycle efficiency drops by 5% from that of the case of no usage of compressed air for cooling of turbine. Assume the following, take properties of gas through turbine as those of air, addition of cooling air to turbine & addition of fuel to compression chamber don't affect turbine power. 0.87 & 0.97 respectively are compression & turbine η .

Ans.

$$T_{inlet_c} = 311 K = T_1$$

$$\frac{P_2}{P_1} = r_p = 8$$

$$\eta_c = 0.87, \eta_t = 0.97$$

$$T_{inlet_t} = T_3 = 1367 K$$

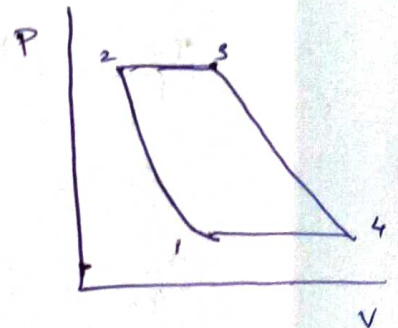
$$P_1 = P_4 = 1 \text{ atm}$$

$$T_2 \text{ \& } T_4 = ?$$

$$\eta_t = \frac{T_3 - T_4}{T_3 - T_{4s}}, \eta_c = \frac{T_2 - T_1}{T_2s - T_1}$$

$$\text{Let } \gamma = 1.4$$

$$\frac{P_2}{P_1} = \left(\frac{T_2}{T_1} \right)^{\frac{\gamma}{\gamma-1}} \Rightarrow 8^{\frac{\gamma-1}{\gamma}} = \frac{T_2s}{T_1}$$



$$\therefore T_{2s} = \underline{563.36}.$$

$$\frac{P_2}{P_1} = \left(\frac{T_3}{T_4} \right)^{\frac{\gamma}{\gamma-1}}$$

$$(8 \times 0.95)^{\frac{0.4}{1.4}} = \frac{1367}{T_{4s}}$$

$$\therefore T_{4s} = \underline{765.79}.$$

$$\left(\frac{P_3}{P_4} \right)^{\frac{0.4}{1.4}} = \frac{T_3}{T_4}.$$

$$\frac{P_3}{P_4} = 0.95 \times \frac{P_2}{P_1}$$

$$\eta_c = \frac{T_{2s} - T_1}{T_2 - T_1} \Rightarrow 0.87 = \frac{563.36 - 311}{T_2 - 311}$$

$$\therefore T_2 = \underline{601 \text{ K}}.$$

$$T_{4s} = 765.79.$$

$$\eta_t = \frac{T_3 - T_4}{T_3 - T_{4s}} \Rightarrow 0.9 = \frac{1367 - T_4}{1367 - 765.7} \Rightarrow T_4 = \underline{826 \text{ K}}.$$

$$W_c = m c_p (T_2 - T_1)$$

$$W_t = m c_p (T_3 - T_4)$$

$$Q = m c_p (T_3 - T_2).$$

$$\eta = \frac{W_t - W_c}{Q} = \frac{m c_p (T_3 - T_4) - m c_p (T_2 - T_1)}{m c_p (T_3 - T_2)}.$$

$$= \frac{T_3 - T_4 - T_2 + T_1}{T_3 - T_2} = \frac{1367 - 826 - 601 + 311}{1367 - 601}$$

$$= \underline{32.7\%} \Rightarrow \text{cycle } \eta \text{ without bleeding.}$$

$$\text{with bleeding, } \eta = 32.7\% - 5\% = \underline{27.7\%}.$$

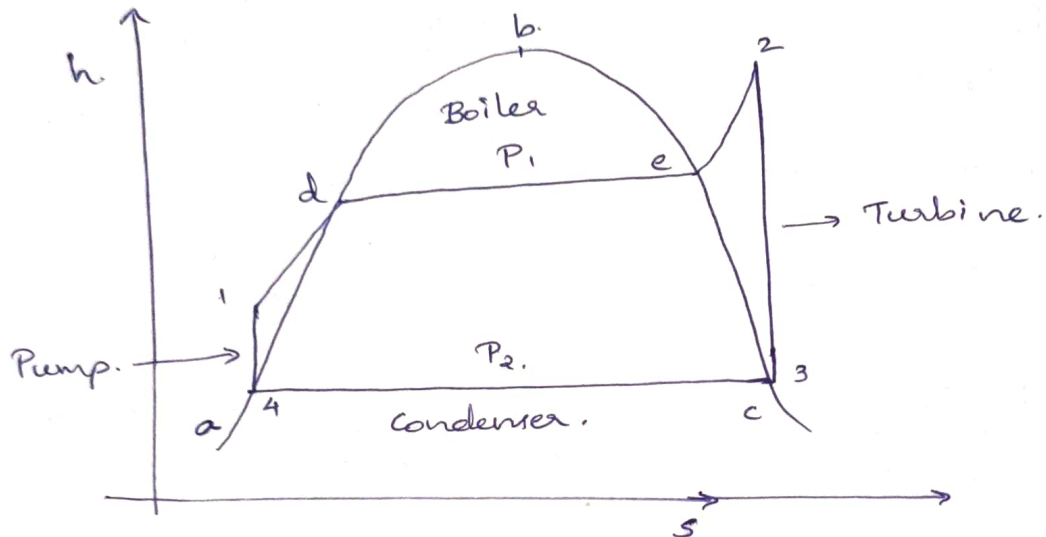
$$0.27 = \frac{m c_p (T_3 - T_4) - (1+x) c_p (T_2 - T_1)}{m c_p (T_3 - T_2)}$$

$$0.27 = \frac{T_3 - T_4 - (1+x)(T_2 - T_1)}{T_3 - T_2}$$

$$\therefore x = \underline{\underline{0.135}}$$

6-11-24.

Rankine Cycle.



1-2 happens in boiler.

e-2 : superheating.

2-c : Expansion in turbine.

c-4 : happens in condenser.

Pulverized coal-fired power plant systems.

1. Coal receipt & preparation.
2. Coal combustion & steam generation.
3. Environmental protection.
4. Turbine generator & electric production.
5. Condenser & feed water system.
6. Heat rejection including cooling tower.